

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10 / CSA1002	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	RAVURU PRANATHI	Reg. No.:	192525076
Date:	22-08-2026	Duration:	60 minutes

Code of Conduct

*I **R.Pranathi(192525076)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: *R. Pranathi*

Annexure A – Research Paper Review / Literature Survey

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **conduct a literature survey and review relevant research papers** to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem. The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.
4. Identify the **limitations and challenges** of the existing research approaches.
5. Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.
6. Discuss the **possible improvements or future research directions** based on the identified research gap.
7. Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper **citations and references** using a consistent referencing style.

Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions & Relevance(15%)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation , Citations & References(15%)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	Mark
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5
Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25

1. Introduction

Construction projects involve several activities such as planning, scheduling, resource management, cost control, progress monitoring, risk management and communication among stakeholders. Traditional construction project management often depends on manual data collection, separate software systems and human decision-making. These practices can result in delays, cost overruns, inefficient resource utilization and difficulty in monitoring project progress.

Recent developments in Artificial Intelligence (AI), Machine Learning (ML), Building Information Modelling (BIM), Digital Twins, Blockchain, IoT and cloud technologies are providing new opportunities to improve construction project management. Recent research shows that AI is increasingly being used for prediction, optimization, decision-making and performance improvement in construction projects.

Therefore, this literature survey reviews recent research on smart technologies for construction project management and identifies the research gap that can be addressed through a **Smart Construction Project Management System**.

2. Objectives of the Literature Survey

The main objectives are:

1. To study recent research on smart construction project management.
2. To identify technologies used in existing construction management systems.
3. To compare AI, ML, BIM and Digital Twin-based approaches.
4. To analyze datasets, tools and methodologies used by researchers.
5. To identify limitations of existing approaches.
6. To determine the research gap.
7. To propose possible improvements for a smart construction project management system.

3. Selected Research Papers

The following **six papers** are suitable for your literature survey.

No.	Year	Research Paper	Main Technology
1	2023	<i>A Blockchain 3.0 Paradigm for Digital Twins in Construction Project Management</i>	Blockchain + Digital Twin
2	2024	<i>Generation of Construction Scheduling through Machine Learning and BIM: A Blueprint</i>	ML + BIM
3	2024	<i>Construction Digital Twin: A Taxonomy and Analysis of the Application-Technology-Data Triad</i>	Digital Twin
4	2024	<i>Digital Twin-Enabled Visibility and Traceability for Building Materials in</i>	5D Digital Twin

		<i>On-Site Fit-Out Construction</i>	
5	2024	<i>Strategic Integration of Drone Technology and Digital Twins for Optimal Construction Project Management</i>	Drone + Digital Twin
6	2025	<i>Artificial Intelligence in Construction Project Management: A Structured Literature Review of Its Evolution in Application and Future Trends</i>	AI

4. Individual Research Paper Reviews

4.1 Zhao, Chen and Xue (2023)

Paper:

“A Blockchain 3.0 Paradigm for Digital Twins in Construction Project Management”

Objective

The research investigates how **Blockchain and Digital Twin technologies** can be combined to improve information management and synchronization in construction project management.

Methodology

The researchers proposed a **Blockchain 3.0-based Digital Twin framework called ChainPM**. Blockchain is used to improve the security and synchronization of information between the physical construction environment and its digital representation.

Technology

- Blockchain
- Digital Twin
- Construction Project Management
- ChainPM

Dataset / Case

The approach was demonstrated using a **pilot modular construction project**.

Major Findings

The study showed that the proposed approach could significantly improve information synchronization. The reported reduction in information synchronization latency was approximately **99.2–99.8%** in the tested scenarios.

Limitation

The approach was demonstrated through a pilot project, so additional validation with different construction projects and larger real-world environments is required.

Relevance

This paper is useful for your project because **secure and synchronized project information** is important in a smart construction management system.

4.2 Al-Sinan, Bubshait and Aljaroudi (2024)

Paper:

“Generation of Construction Scheduling through Machine Learning and BIM: A Blueprint”

Objective

The objective is to develop a blueprint for automatically generating construction schedules using **Machine Learning and BIM**.

Methodology

The proposed system uses previous **BIM IFC 3D files** and schedules prepared by experienced construction schedulers to train an ML model.

The model can potentially determine:

- Construction activities
- Activity sequence
- Duration
- Cost
- Required resources

The authors aim to reduce dependence on manually developed scheduling constraints.

Technology

- Machine Learning
- BIM
- IFC
- Autonomous scheduling

Dataset

Previous construction projects containing **BIM IFC files and expert-created schedules**.

Major Findings

The research demonstrates the potential for ML to automatically generate construction schedules and estimate activity duration, cost and resources.

Limitation

The paper presents a **blueprint rather than a fully validated production system**. The authors identify the need for implementation using different project types and comprehensive evaluation.

Relevance

This is highly relevant to your proposed system because your system can include **AI-based scheduling and task management**.

4.3 Saif, RazaviAlavi and Kassem (2024)

Paper:

“Construction Digital Twin: A Taxonomy and Analysis of the Application-Technology-Data Triad”

Objective

The research aims to understand how **Digital Twin technology** can be applied during the construction phase.

Methodology

The researchers developed a taxonomy of Digital Twin applications and analyzed the relationship between:

Application → Technology → Data

They identified **seven application areas and 19 uses** for construction Digital Twins.

Technology

- Digital Twin
- BIM
- Construction data
- Data integration

Dataset

The research is primarily a **literature/review-based study** rather than a conventional machine-learning dataset.

Major Findings

The study proposes a conceptual **five-layer Digital Twin architecture** and identifies important relationships between construction applications, technologies and data.

Limitation

Integration of different technologies and data sources remains challenging. The study also identifies gaps requiring further research.

Relevance

This paper supports the use of a **Digital Twin or digital project representation** in your proposed Smart Construction Project Management System.

4.4 Yang, Li, Yu and Zhong (2024)

Paper:

“Digital Twin-Enabled Visibility and Traceability for Building Materials in On-Site Fit-Out Construction”

Objective

The study aims to improve the **visibility and traceability of construction materials**.

Methodology

The researchers proposed a service-oriented **5D Digital Twin** that connects physical construction activities with digital information.

The system uses:

- AutomationML
- OPC UA
- Digital Twin
- Real-time data

Dataset / Case

The approach was validated using a **practical construction case study**.

Major Findings

The proposed Digital Twin provides real-time visibility and traceability of building materials and can support more efficient construction management.

Limitation

The research focuses mainly on **material visibility and traceability**, rather than providing a complete project management solution.

Relevance

Your system can extend this concept by including **resource/material tracking together with scheduling, cost and progress monitoring**.

4.5 Strategic Integration of Drone Technology and Digital Twins (2024)

Paper:

“Strategic Integration of Drone Technology and Digital Twins for Optimal Construction Project Management”

Objective

The research investigates the integration of **drones and Digital Twins** for construction project monitoring.

Methodology

Drones are used to collect construction-site images/data, while image-processing technologies and Digital Twins are used to update project information.

Technology

- Drones
- Image processing
- Digital Twin
- Construction monitoring

Dataset / Case

Construction-site monitoring data and images.

Major Findings

The approach can provide project managers with updated information about construction progress and support

better decision-making. It can also improve communication and coordination among project stakeholders.

Limitation

The approach requires suitable drone infrastructure, image-processing capabilities and integration with project-management systems.

Relevance

Drone-based progress monitoring could be added as a future module to your proposed system.

4.6 Adebayo et al. (2025)

Paper:

“Artificial Intelligence in Construction Project Management: A Structured Literature Review of Its Evolution in Application and Future Trends”

Objective

The study investigates the evolution of **AI applications in construction project management**, including benefits, challenges and future research directions.

Methodology

The researchers conducted a structured literature review using databases including:

- Scopus
- Web of Science
- ProQuest

They synthesized **135 peer-reviewed articles** published across different periods.

Technology

- Artificial Intelligence
- Machine Learning
- Predictive analytics
- Optimization
- Intelligent decision-making

Major Findings

The research found that AI is particularly applied to:

- Planning
- Monitoring
- Control
- Prediction
- Optimization
- Decision-making
- Performance improvement

Limitation

The study identifies continuing challenges related to **data availability, implementation, skills, organizational readiness and practical adoption.**

Relevance

This paper provides strong overall support for developing an **AI-enabled Smart Construction Project Management System.**

5. Comparative Literature Survey

Paper	Year	Methodology	Technology	Dataset / Case	Main Finding	Limitation
Zhao et al.	2023	Blockchain-DT framework	Blockchain + DT	Modular construction pilot	Improved information synchronization	Limited pilot validation
Al-Sinan et al.	2024	ML-based scheduling blueprint	ML + BIM	BIM IFC + expert schedules	Automatic scheduling, cost and resource estimation	Requires full implementation and validation
Saif et al.	2024	Literature-based taxonomy	Digital Twin	Research literature	7 application areas and 19 uses identified	Integration/data challenges
Yang et al.	2024	5D DT framework	Digital Twin + AutomationML + OPC UA	Construction case	Real-time material visibility and traceability	Focuses mainly on materials
Drone + DT study	2024	Drone monitoring + image processing	Drone + DT	Construction-site data	Improved progress monitoring and communication	Infrastructure/integration requirements
Adebayo et al.	2025	Structured literature review	AI/ML	135 peer-reviewed papers	AI improves prediction, optimization and decision-making	Practical adoption/data challenges

The comparison shows that existing research generally focuses on **one or two specific functions**, such as scheduling, material tracking, information synchronization or site monitoring, rather than providing one simple platform covering the complete project-management lifecycle.

6. Comparative Analysis

The existing approaches can be compared based on important construction project-management requirements.

Parameter	Blockchain + DT	ML + BIM	Digital Twin	Drone + DT	AI-based CPM
Project Scheduling	Medium	High	Medium	Low	High
Resource Management	Medium	High	High	Low	High
Progress Monitoring	Medium	Medium	High	High	High
Material Tracking	High	Medium	High	Medium	Medium
Prediction	Medium	High	Medium	Medium	High
Data Security	High	Medium	Medium	Medium	Medium
Real-time Monitoring	High	Medium	High	High	High
Complete Project Management	Low–Medium	Medium	Medium	Low–Medium	Medium
Ease of Integration	Medium	Medium	Low–Medium	Medium	Medium

Analysis

The literature indicates that **ML+BIM is particularly useful for automated scheduling**, Digital Twins are useful for **real-time representation and monitoring**, drones can improve **site-progress monitoring**, and Blockchain can improve **information security and synchronization**.

However, no single approach among the selected studies provides a simple integrated system covering **tasks + scheduling + resources + progress + cost + risk + intelligent decision support**.

7. Limitations of Existing Research

The following limitations were identified:

1. Fragmented solutions

Most studies address a specific problem such as scheduling, material tracking or monitoring rather than complete construction project management.

2. Data availability

AI and ML models require sufficient historical construction data. Limited availability of reliable real-world datasets can affect model performance.

3. Integration problems

BIM, Digital Twin, AI, IoT and other technologies often use different data formats and platforms.

4. High implementation cost

Advanced Digital Twin, drone and IoT systems may require significant hardware, software and infrastructure.

5. Lack of real-world validation

Several proposed frameworks and blueprints still require testing across different construction projects and environments.

6. Skill requirements

Construction organizations may lack employees with sufficient expertise in AI, BIM, data analytics and Digital Twin technologies.

7. Data security

Construction projects contain sensitive information, making cybersecurity and access control important.

8. Identified Research Gap

Major Research Gap

Based on the reviewed literature, there is a gap in developing a **simple, integrated and user-friendly Smart Construction Project Management System** that combines multiple project-management functions in one platform.

Existing research mainly focuses on individual technologies:

ML + BIM → Scheduling

Digital Twin → Monitoring

Drone + DT → Site Progress

Blockchain + DT → Secure Information

AI → Prediction and Decision Support

However, these capabilities are generally investigated separately.

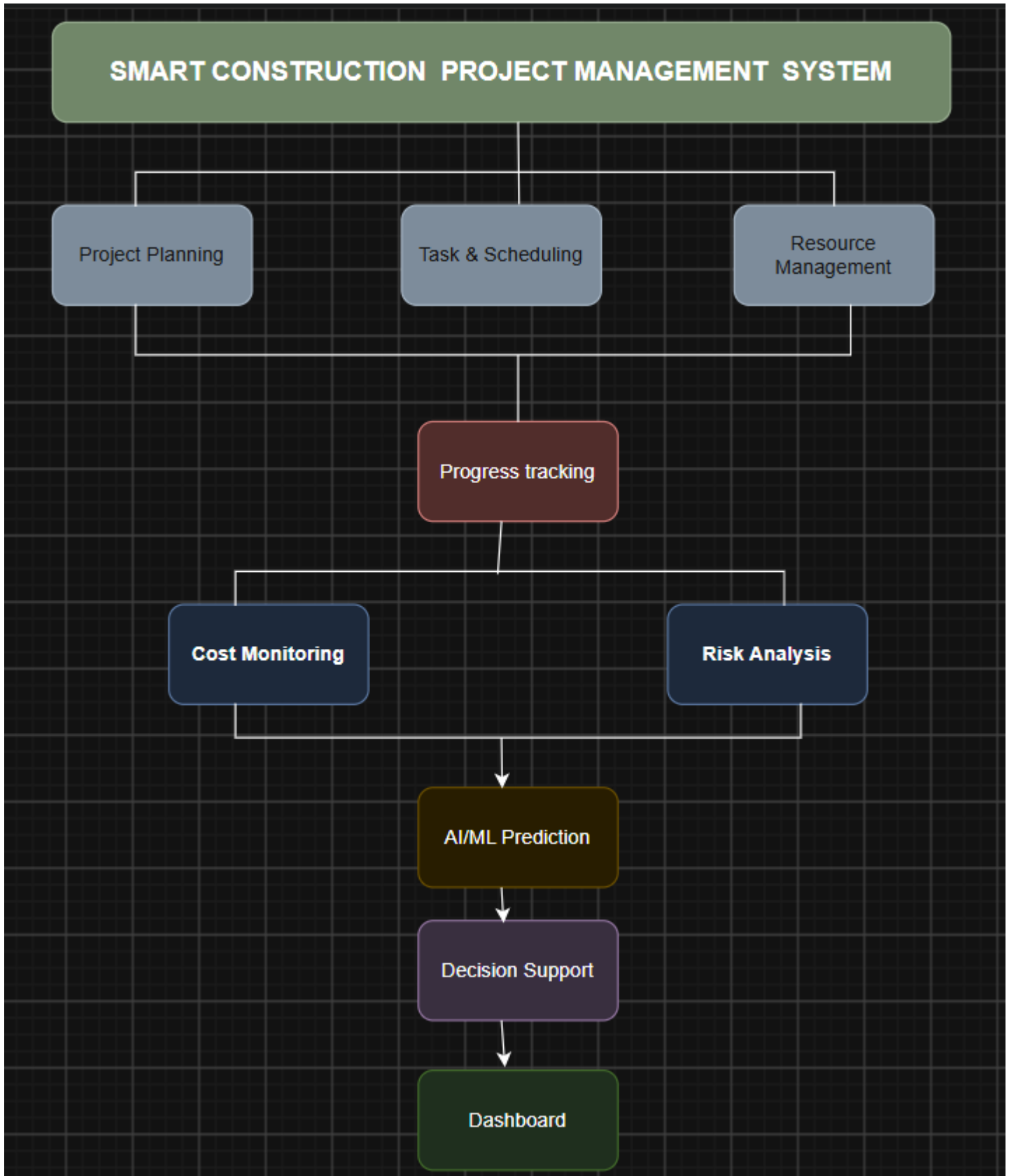
Proposed Research Gap Statement

There is a need for an integrated Smart Construction Project Management System that combines project scheduling, task management, resource monitoring, progress tracking, cost management and AI-based prediction within a single platform, while remaining scalable, secure and easy to use.

This gap is also consistent with recent literature identifying incomplete automation, data limitations and the need for more comprehensive AI applications across construction project-management functions.

9. Proposed Smart Construction Project Management System

Based on the identified gap, the proposed system can contain the following modules:



10. Possible Future Research Directions

Based on the identified research gap, the following improvements can be considered:

1. AI-based delay prediction

Use historical project data to predict possible project delays.

2. Automated scheduling

Integrate ML with BIM to automatically generate and update project schedules.

3. Resource optimization

Use AI to predict required manpower, equipment and materials.

4. Digital Twin integration

Create a digital representation of the construction project and continuously update it using real-time data.

5. IoT integration

Use sensors to monitor:

- Equipment
- Materials
- Environmental conditions
- Worker/site conditions
- Construction progress

6. Drone-based monitoring

Use drones and computer vision to automatically identify construction progress.

7. Cost prediction

Develop ML models for estimating and monitoring project costs.

8. Secure data management

Blockchain or secure cloud architecture can be investigated for protecting project information.

9. Mobile application

Provide site engineers and project managers with real-time access through a mobile application.

10. Integrated dashboard

Provide a single dashboard displaying:

Schedule + Cost + Resources + Progress + Risks + Alerts

11. Conclusion

The literature survey demonstrates that smart technologies are increasingly being applied to construction project management. AI and ML provide opportunities for prediction, optimization and automated scheduling, while BIM provides structured project information. Digital Twins enable real-time representation and monitoring, drones can improve site-progress monitoring, and Blockchain can improve information security and synchronization.

However, existing studies are largely focused on individual functions or specific technologies. There remains a need for an integrated